

Ziyang ZHAN

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EDUCATION

University of Illinois at Urbana-Champaign
Master of Computer Science

Champaign, IL
May 2021 -- August 2023

University of South Carolina
Master of Hospitality and Tourism Management

Columbia, SC
January 2019 -- December 2020

Jiangxi Normal University
Bachelor of Software Engineering

Jiangxi, China
September 2011 -- May 2015

HONORS

Phi Kappa Phi Honor (Top 5% of Graduate Students)
Eta Sigma Delta Honor (Top 5% of Graduate Students)
Outstanding Graduates Awards

UIUC, 2022
USC, 2020
JNU, 2015

SKILLS

Frontend: Excellent Flutter, React, PySide6
Backend: Node.js, Go, MongoDB, SQL
Cloud: AWS
Language: Proficient English, Native Mandarin, Entry-level Cantonese

RELATED EXPERIENCE

[Hong Kong Applied Science and Technology Research Institute](#)

HK, Nov 2023 — Present

LVLM-RPA System—Full Stack Engineer

- Built a **PySide6** screen recorder capturing video and user actions across Windows apps (**Word, Excel, Chrome, Edge, Firefox**) with timestamped event logs.
- Integrated **Playwright** to align video frames with UI actions and DOM selectors, generating paired training data (**frame + action sequence**) for VLM fine-tuning.

OCR Recognition Platform—Full Stack Engineer

- OCR Model (PyTorch + FastAPI)**: Fine-tuned TrOCR with a synthetic image data generator server (diverse fonts + noise injection) to improve real-world generalization without manual labelling
- Frontend (Flutter)**: Built responsive Flutter app targeting Web + iOS with MVVM (Provider); implemented caching strategies and WebSocket for real-time OCR status updates.
- Backend (Node.js + MongoDB)**: Designed **RESTful APIs** with chunked uploads for large PDFs, Bull queue for async OCR job scheduling, and worker pool for parallel page processing—achieving 1,000+ docs/hour with 99.9% uptime and 35% faster queries.

Comprehensive Federated Learning Platform

- As the first project at the company, rapidly onboarded **Flutter with MVVM** architecture (Provider) and **MongoDB** to design and deliver a responsive model monitoring dashboard
- Designed and implemented **unit and integration tests**, utilizing dependency injections and mocking to verify page navigation and widget states
- Drafted the whitepaper: [Federated Learning: Unlocking Innovation in the Insurance Sector](#)

Hong Kong Polytechnic University

Social Listening Platform in Greater Bay Area (**TOUROMETER**)

Hong Kong
February 2022 – May 2023

- Model Development**: Built SOTA **multi-lingual** (Simplified/Traditional Chinese, Japanese, Korean, and English) **sentiment models** (CNN, Bi-LSTM) with >90% accuracy and a topic-relevance classifier for tourism relevance
- Real-time Data Scraping System**: Built a **24/7 scraping engine** collecting >10T real-time data from major social media/news platforms (Weibo, Twitter, Facebook, etc.) with **anti-blocking mechanisms** including dynamic IP rotation and human behavior simulation

- **Product Deployment:** Created an interactive **vanilla JS + D3.js dashboard** delivering real-time tourism insights and policy recommendations to government stakeholders

University of Illinois at Urbana-Champaign

Remote

Next-Generation Search Engine — UIUC [Forward Data Lab](#)

January 2023 – August 2023

- **Model Development:** Fine-tuned **NER models (BERT/RobERTa)** for search query understanding.
- **Data Pipeline:** Built an automated data labeling generator using external knowledge bases (Wikipedia/Frebase) to produce large-scale training data from unannotated search logs.
- **Product Deployment:** Deployed the model as a **FastAPI** service on **AWS** and developed a **Manifest Chrome Extension** that highlights entities instantly upon user text selection.

ADDITIONAL EXPERIENCE

University of South Carolina

Columbia, SC

Academic Tutor

September 2019 – December 2020

- Assisted >50 international students in Quantitative Methods course, and 85% of students received an A
- Help international students adapt to the academic course and improve class performance. So, the students are more confident to complete the master's degree